温州肯恩大学
WENZHOU-KEAN UNIVERSITY

GMBA 5885 W01
Strategic Mgt Global Info Sys

Strategic Use of Information Resources

Dawei Chen

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Learning Objectives

By the end of this session, you should be able to:

- Understand what **information resources** are in a firm
- Explain how MIS can shape *competition*, *value creation*, and *capability*
- Apply Five Forces, Value Chain, and Resource-Based View to analyze a firm's MIS
- Evaluate whether MIS creates operational support, temporary advantage, or sustained advantage

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Opening Question

Why do some firms gain advantage from MIS, while others just spend money on technology?

Firm A

Same ERP / CRM / cloud system

- ➕ Faster decisions and response times
- ➕ Better coordination across units
- ➕ Clearer visibility into operations

Firm B

Same ERP / CRM / cloud system

- ➖ Employees revert to old workflows
- ➖ Data exists but is not acted on
- ➖ Investment does not create value

What makes the difference?

Opening Case: Zara

1. What makes Zara's use of information systems strategic rather than just operational?
2. How does information flow – from customers to designers to production – create value for Zara?
3. Why can't competitors like H&M or Gap easily copy Zara's system or speed?
4. What general lessons can managers learn from Zara about using information to compete and stay ahead?

Today's Logic

Competition → Activities → Capabilities

Lens 1 – Five Forces

Where is the pressure?

Identifies the competitive forces shaping the industry and shows how MIS can shift them in the firm's favor

Outside the firm

Lens 2 – Value Chain

Where does MIS add value?

Maps the activities where MIS improves efficiency, coordination, or customer experience inside the firm.

Inside the firm

Lens 3 – RBV

Does the advantage last?

Asks whether the MIS capability is valuable, rare, and hard enough to imitate to create sustainable advantage.

Sustaining the firm

Information Resource

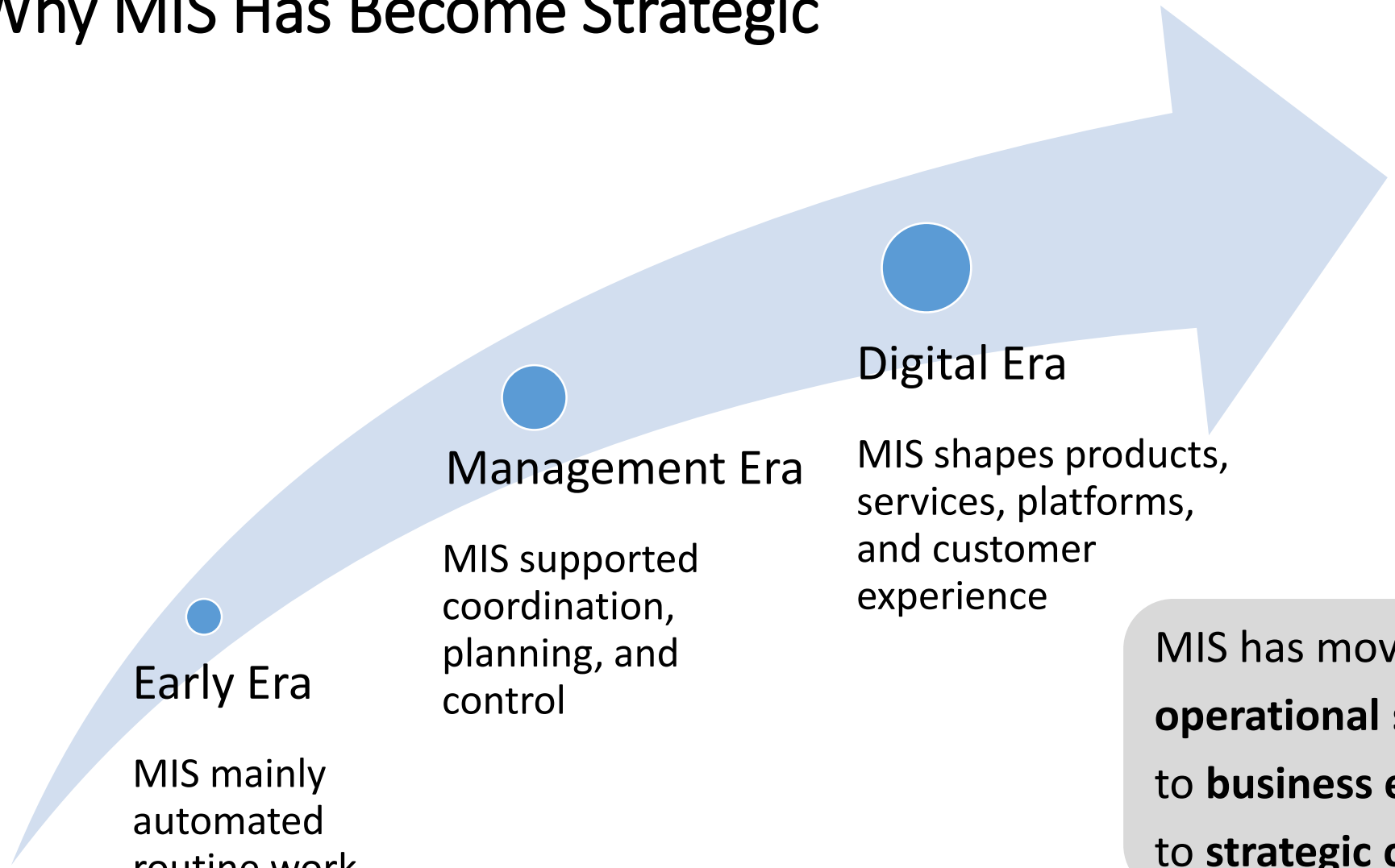
Information resources are the **data**, **technology**, **people**, and **processes** that allows a firm to sense, decide, coordinate, and act.

Examples:

- Customer and operational data
- ERP, CRM, SCM, digital platforms
- Analytics tools and dashboards
- IT staff and business users
- Standardized business processes
- Vendor and partner connections

MIS is not only technology — it is the organized use of information to support and shape business performance.

Why MIS Has Become Strategic



MIS has moved from **operational support** to **business enablement** to **strategic co-creation**

IT Assets vs. IT Capabilities

Information Resources

IT Assets

- *IT infrastructure* – Share digital foundation for operations and delivery
- *Information repository* – Shared data foundation for storage and use

IT Capabilities

- *Technical skills* – Build and implement information systems
- *Management skills* – Plan and manage IT resources and projects
- *Relationship skills* – Align IT with business and partners

<u>Assets can be purchased. Capabilities are built over time.</u>

Competition and Pressure

Generated by ChatGPT Thinking

1) What pressure does the firm face?

2) How can MIS help it respond?

3) Does MIS create advantage, or only ***help the firm survive***?

Porter’s Five Forces: Why It Matters for MIS

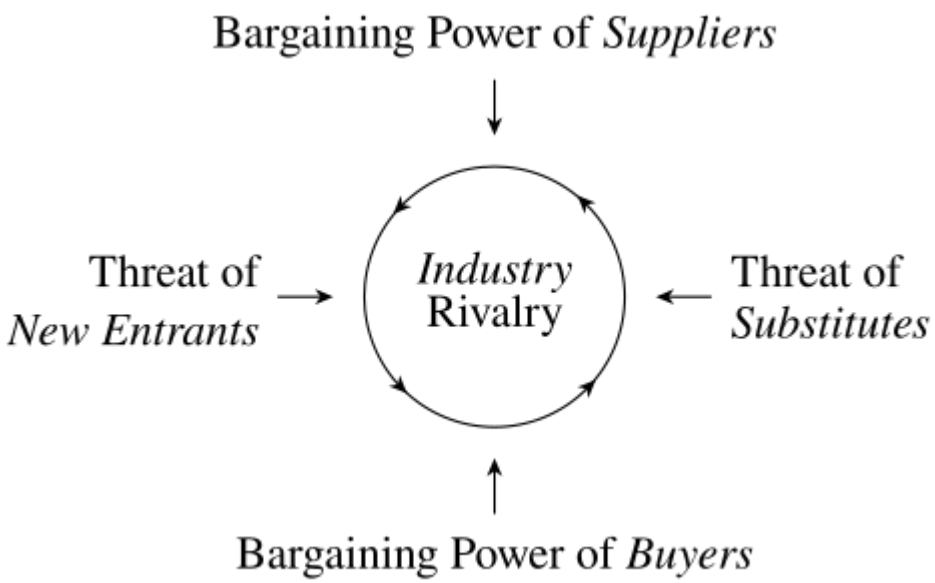
Five Forces helps us ask:

- Where is the pressure?
- Which force hurts profitability most?
- How can MIS reduce that pressure or create leverage?

MIS can influence competition through:

- speed and coordination
- personalization
- switching costs
- supply chain visibility
- data advantages
- new business models

Porter's Five Forces 1979



Source: Wikipedia

Michael Porter

- A framework for analyzing a firm's competitive environment.
- It explains how **five forces** shape industry *profitability* and *attractiveness*.

MIS Through Five Forces Lens

Force	How MIS Can Respond
Competitive rivalry	improve pricing, analytics, and speed of response
Threat of new entrants	create switching costs, ecosystems, and data advantages
Threat of substitutes	improve digital experience, integration, and convenience
Buyer power	support CRM, loyalty systems, and customization
Supplier power	improve SCM visibility, data sharing, and coordination

Zara Through the Five Forces Lens

Analysis

- Fashion retail is highly competitive; speed and trend responsiveness matter
- Retail entry is possible, but fast-response execution is harder to build
- Customers have many alternatives; speed and design reduce substitute pressure
- Information sharing and coordination reduce supplier dependence

Discussion

1) Which force is **strongest**?
2) How does **MIS** help Zara respond?
3) Is MIS creating advantage, or just helping Zara keep up?

From Pressure to Value Creation

Five Forces

Where is the pressure?

Identifies the external forces shaping competition — rivals, buyers, suppliers, substitutes, new entrants.

Value Chain

Where does the firm respond?

Traces the internal activities where MIS helps the firm act on that pressure — faster, cheaper, or better.

MIS becomes strategic when it helps the firm respond to real competitive pressure, not simply because technology is present.

Value Creation and Capability

Value Chain

External pressure

RBV

Internal value creation

Sustainable advantage

Key Questions

- Where does MIS add value inside the firm?
- When does that value become hard to copy?

Value Chain: Why It Matters for MIS

Value Chain helps us ask:

- Where does MIS improve performance?
- Which activities benefit most from data and system integration?
- How does MIS connect activities across the firm?

MIS adds value through:

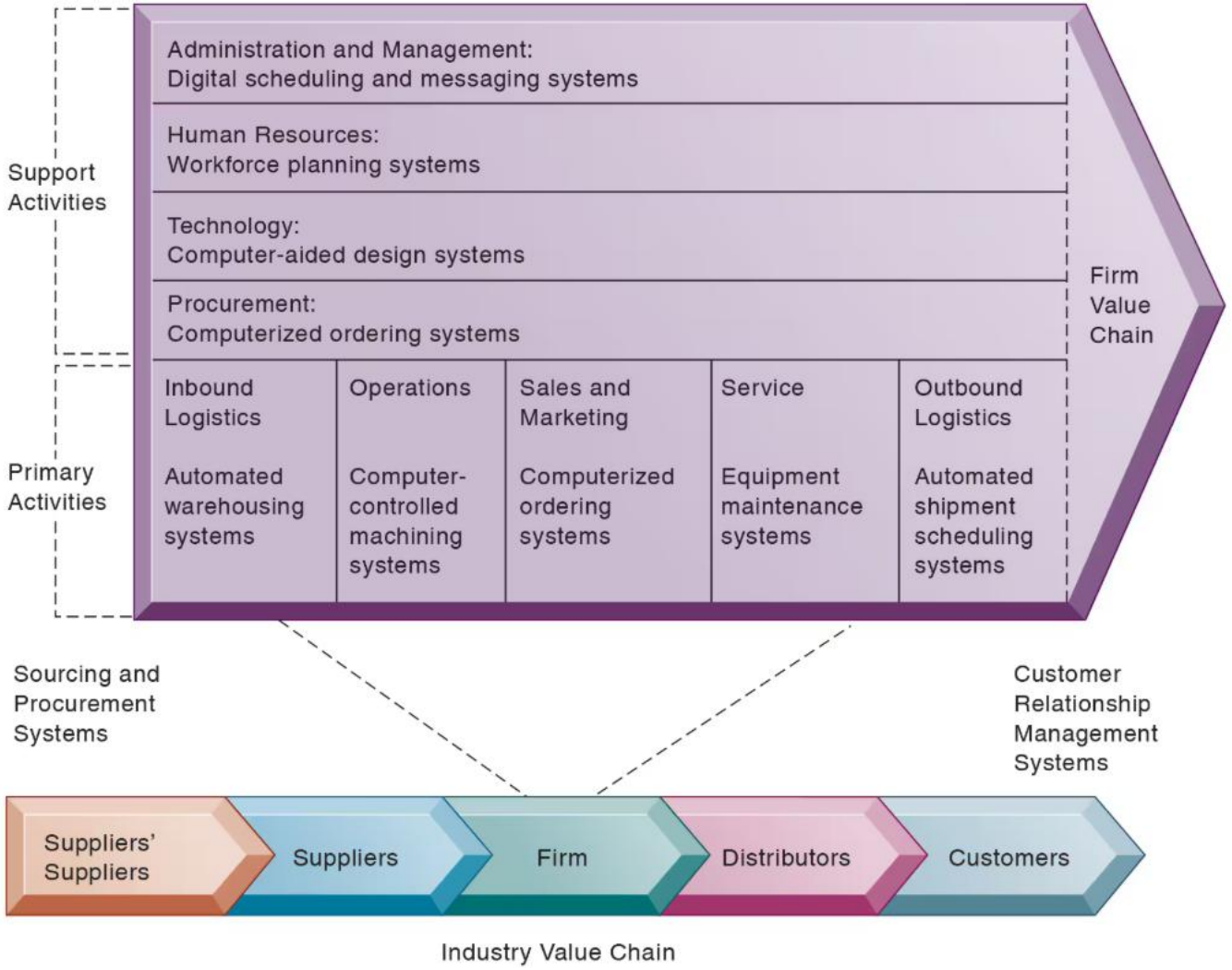
- Visibility
- Speed
- Coordination
- Consistency
- Customization
- lower error and transaction cost

Value Chain Model

Primary activities directly create customer value through core processes:

→ Inbound logistics, operations, outbound logistics, sales & marketing, and service.

(From receiving materials → producing goods → delivering → selling → maintaining products.)



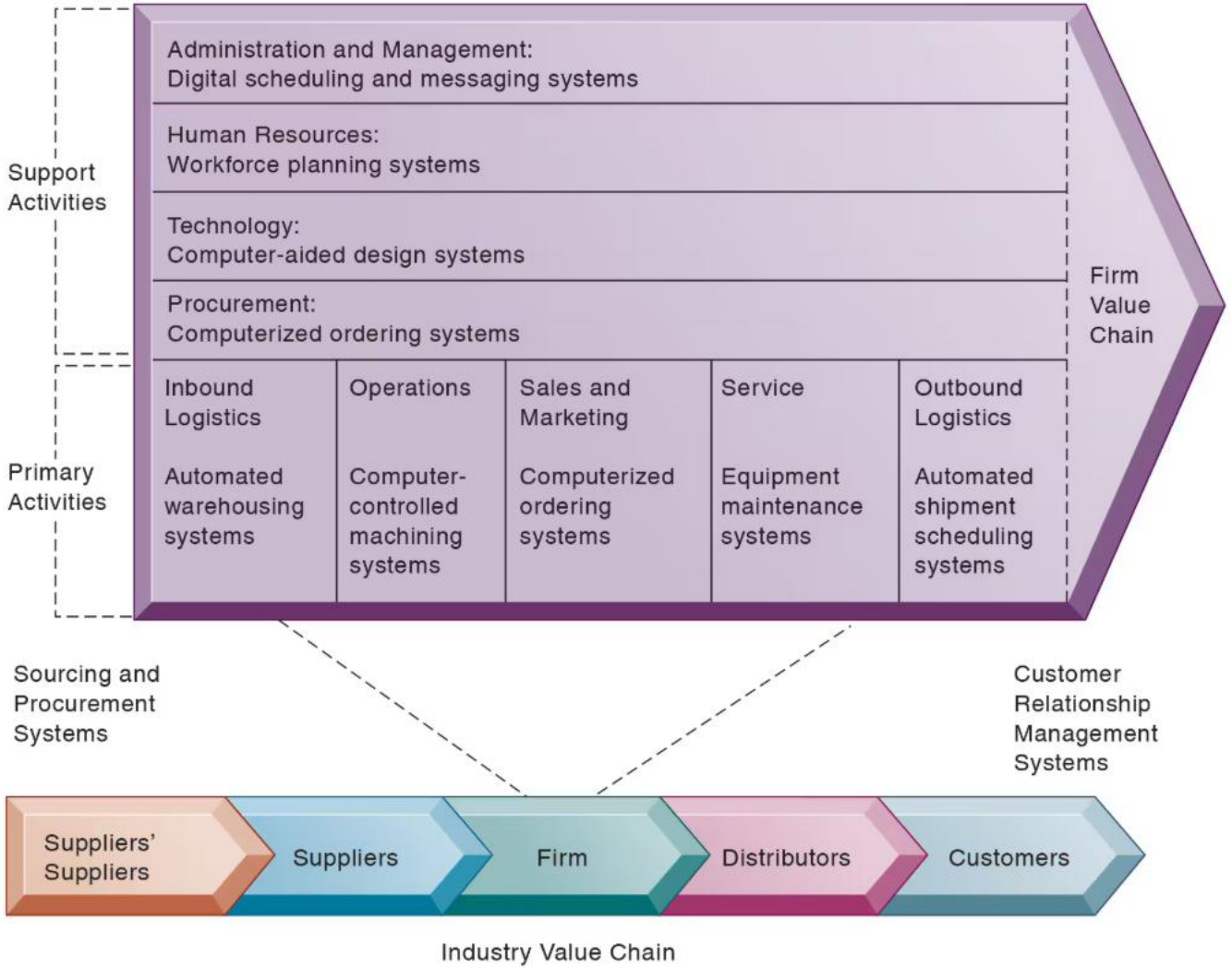
Laudon et al. 18th

Value Chain Model

Support activities enable and enhance primary activities through:

→ Infrastructure, human resources, technology development, and procurement.

(They provide the foundation that allows core operations to run efficiently.)



Laudon et al. 18th

MIS in the Value Chain

- Inbound logistics → inventory visibility, demand sensing
- Operations → ERP, workflow systems, automation
- Outbound logistics → routing, tracking, fulfillment
- Marketing and sales → CRM, recommendation systems, pricing tools
- Service → ticketing systems, self-service, AI support
- Support activities → HRIS, supplier portals, analytics, integrated reporting

Zara Through the Value Chain Lens

Activity	Zara's Value Chain
Inbound logistics	Better visibility into inventory and demand
Operations	Rapid planning and adjustment of production
Outbound logistics	Frequent replenishment and fast response
Marketing and Sales	Store data signals customer preference
Support activities	Systems connect design, production, logistics, and stores

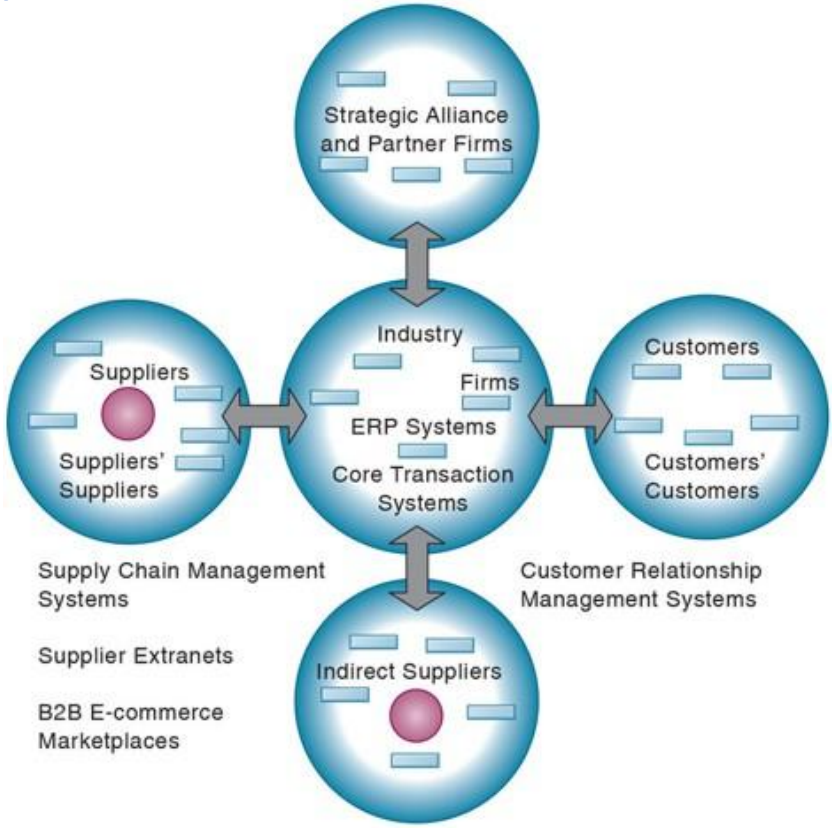
Where does MIS add the **most value** for Zara?

The Value Web

From internal efficiency to inter-firm coordination

The Value Web

- A network of independent firms using IT to coordinate their value chains to produce a product or service for a market collectively.
- Non-linear, customer-driven, flexible, and adaptive to market changes.
- Enables faster response, time-to-market, and collaborative innovation.



Laudon et al. 18th

From Value to Advantage

Value Chain

Where does the firm respond?

Traces the internal activities where MIS helps the firm act on that pressure — faster, cheaper, or better.

RBV

Whether competitors can copy?

Identifies the internal resources that creates sustained competitive advantage

RBV: Why It Matters for MIS

Owning technology is easy. **Building advantage from it is harder.**

RBV helps us ask:

- Is this information resource valuable?
- Is it rare?
- Is it difficult to imitate?
- Is it difficult to substitute?
- Is the firm organized to use it well?

A standard system rarely creates lasting advantage by itself.

Resource-Based View

Evaluate whether a firm's information resources attain and sustain competitive advantage

To **Attain** Advantage	To **Sustain** Advantage
Valuable & Rare → improve efficiency, effectiveness, innovation	Hard to Imitate, Substitute, or Transfer → embedded in processes, relationships, culture
Example: Early mobile banking apps created value but soon became "table stakes."	*Example:* Walmart's logistics IS or firm-specific analytics routines are hard to copy.

<u>Many systems help a firm perform better.</u>

<u>Far fewer create durable advantage.</u>

What Is Easy to Copy – and What Is Not?

Easier to Copy

- Standard ERP
- Common cloud tools
- Basic dashboards
- Off-the-shelf analytics

Harder to Copy

- Proprietary data
- Cross-functional routines
- Strong IT-business collaboration
- Fast decision loops
- Culture of using data in operations

The MIS advantage is often not the system itself, but how the organization uses it.

Sustainability Barriers

Barrier	Description
IT project barrier	The initiative is difficult, costly, or time-consuming to replicate
IT assets and capabilities barrier	The firm has data, skills, or integration capability others lack
Complementary resources barrier	Advantage depends on brand, process, supplier network, or culture in addition to IT
Preemption barrier	The firm moves first and locks in customers, partners, or data

Piccoli, G., & Ives, B. (2005). IT-dependent strategic initiatives and sustained competitive advantage: A review and synthesis of the literature. MIS Quarterly, 747-776.

How Innovation Sustain Advantage

Even the strongest advantages erode.
Sustainability requires continuous reinvention — not stability.

1. Enhance Value with IT

- Improve products, services, and customer experience
- Use data and AI to continuously refine offerings

2. Anticipate Disruption

- Act before competitors or new entrants
- Compete with your future self, not just rivals

3. Balance Horizons

- Short-term: incremental improvements
- Long-term: business model transformation

Example: Netflix

- DVD rental → Streaming platform → Content producer
- Proactively cannibalized its own model
- Accepted short-term losses to lead industry shift

→ Turned risk into long-term dominance

Zara Through the RBV Lens

Questions

- Is Zara's MIS valuable?
- Is it rare?
- What exactly is hard to imitate?

Focus

- Not just software
- Fast data-to-decision process
- Tight coordination among stores, design, and operations
- Embedded routines and organizational discipline

The real MIS advantage often comes from integration and use, not ownership alone.

Zara Through the RBV Lens

	Value Creation		*Value Sustainability*		
Resource/Attribute	*Value*	*Rarity*	*Imitation*	*Substitution*	*Transfer*
IT Infrastructure	**Moderate –** effective use of POS, handhelds, and automation, but technologies are replicable by competitors		**Easy** to imitate and transfer its infrastructure **Moderate** for substitution of infrastructure (automated conveyers)		
Information Repository	**High –** valuable and rare customer data, tightly integrated with operations and design decisions		**Difficult** to imitate and transfer **Extremely difficult** to substitute because of the volume and nature of the data		

Pearlson et al. 2024

Zara Through the RBV Lens

	Value Creation		*Value Sustainability*		
Resource/Attribute	*Value*	*Rarity*	*Imitation*	*Substitution*	*Transfer*
Technical Skills	**Low —** limited value/rarity; technical talent is widely available		**Moderately difficult** to imitate or substitute; offers some sustainability through cross-system integration		
IT Management Skills	**High —** valuable and rare, accumulated over time		**Difficult** to imitate, substitute, or transfer; resources leveraged well		
Relationship Skills — *Externally Focused*	**High value –** strong supplier relationships **Moderate rarity –** competitors can also built relationships with suppliers		**Difficult** to imitate, substitute, or transfer – ~5-week end-to-end turnaround		
Relationship Skills — *Spanning*	**High** rarity of spanning		**Difficult** to imitate, substitute, or transfer – tight-knit teams enable fast, accurate responses to customer needs		

Pearlson et al. 2024

Risks of Competing Through MIS

Implementation Risks

- Poor adoption
- Weak process redesign
- Cost overruns
- System complexity

Competitive Risks

- Easy imitation
- Expensive technology races
- Overdependence on one vendor or platform

Legal Risks

- Privacy and security exposure
- Compliance failures
- Data misuse

MIS can create advantage, but also cost, complexity, and vulnerability.

Managerial Takeaway

In many information-intensive businesses, business strategy and IS strategy are no longer separate.
They must be increasingly co-created.

- MIS is not just infrastructure
- It shapes products, services, processes, and decisions
- Managers must ask not only “What system do we need?”
- But also “What business can we become because of this system?”

Synthesis

Lens 1 – Five Forces

Where is the pressure?

Identifies the competitive forces shaping the industry and shows how MIS can shift them in the firm's favor

How MIS responds to competition

Lens 2 – Value Chain

Where does MIS add value?

Maps the activities where MIS improves efficiency, coordination, or customer experience inside the firm.

Where MIS adds value

Lens 3 – RBV

Does the advantage last?

Asks whether the MIS capability is valuable, rare, and hard enough to imitate to create sustainable advantage.

Whether MIS creates sustainable advantage

Does this MIS merely support operations, or does it change how the firm competes?

Apply to Your Project

Discuss:

- What are the key information resources in your analyzed company?
- Where does MIS add the most value?
- Is the MIS mainly:
  - infrastructure?
  - an operational tool?
  - a real capability?
- What IS-related problem can you identify?

Reference & AI Use Disclosure

- Reference
  - Pearlson, K. E., Saunders, C. S., & Galletta, D. F. (2024). *Strategic use of information resources*. In *Managing and using information systems: A strategic approach* (8th ed.). Wiley.
  - Laudon, K. C., Laudon, J. P., & Traver, C. G. (2025). *Information systems, organization and strategy*. In *Management information systems: Managing the digital firm* (18th ed.). Pearson.
- How AI was used
  - AI was used as a thinking and design assistant to:
    - redesign session flow
    - simplify textbook-heavy content
    - develop case-based discussion prompts
    - strengthen links between MIS, strategy, and the group project
- Instructor role
  - All content was reviewed, selected, and adapted by the instructor for this course context.

Case: Zhuiyi Technology

1) (Information resources) What are the key information resources Zhuiyi relies on? Which are assets, and which are capabilities?

2) (Five Forces) What are the strongest competitive pressures? How does MIS help Zhuiyi respond?

3) (Value Chain) Where exactly does MIS add value? Which activities are most affected?

4) (RBV) What part of Zhuiyi's MIS is valuable, rare, and hard to imitate? Is the advantage sustainable?

5) Should Zhuiyi invest more in new AI products or in deeper industry customization / diversification? Why? What would you suggest Joey Wu do for the next 3-5 years?